Mnemonic Similarity Task

& Event Segmentation

N = 160 participants

3 conditions: Item-Only · Task-Only · Both Item+Task

160	24,000	5	3
Participants	Test Trials	Lure Bins	Event Positions

01 Introduction & Background

The domain, the task, and why it matters

Replication of Morse et al. (2023)

The Hippocampus & Pattern Separation

Why similar memories don't interfere with each other

Pattern Separation

Hippocampal process that orthogonalises similar inputs into non-overlapping neural representations

Pattern Completion

Retrieving a full memory from a partial cue, the complementary process to pattern separation

MST Paradigm

Behavioural proxy for hippocampal pattern separation using perceptually similar lure images

Mnemonic Similarity Task — Design

Replication of Morse et al. (2023) with event boundary manipulation

Encoding

280 images
40 events × 7 items
"Fits in a shoebox?"
(f/j)

→

Test

150 items
60 targets + 60 lures
+ 30 foils

→

Response

O = Old
S = Similar
N = New

Lure Similarity Bins:

1 — Most Similar	2	3	4	5 — Least Similar

Event Segmentation Theory (EST)

Zacks et al., 2007 · Morse et al., 2023

Core Idea

- Experience is continuously parsed into discrete events
- Event boundaries occur when predictions fail
- Boundaries trigger a cognitive reset → reallocation of processing resources
- Post-boundary items may benefit from enhanced encoding
- Response times should increase at boundaries

Position × Memory Prediction

Pre-boundary	Standard encoding; pre-boundary context active
Mid-event	Deep within event; stable event model; best context
Post-boundary	After boundary; fresh encoding start; episodic reset

02 Hypotheses & Methods

Research questions, planned analyses, and statistical approach

Research Hypotheses

Confirmatory (H1–H4) and Exploratory (H5–H7)

H1 Memory above chance: REC and LDI > 0 for all groups (verified) ✓

H2 Boundary benefit: Post-boundary items show higher REC/LDI than mid-event

H3 Lure gradient: LDI lower for Bin 1 (most similar) vs Bin 5 (verified) ✓

H4 Group differences: Both/Task-Only differ from Item-Only in memory

H5 Exploratory: Longer RT at test for post-boundary/mid-event items

H6 Exploratory: Positive correlation between REC and LDI within-subjects

H7 Exploratory: Position × lure bin interaction on LDI

Methods: Data Processing Pipeline

320 CSV files ⟶ per-participant REC and LDI

1

Load CSVs

320 files across 3 groups
Participant ID = first 5 digits

→

2

Classify

a.jpg = Target
b.jpg = Lure
Foils/ = Foil

→

3

Assign Bins

Merge Set6/SetScC bins.txt
Bin 1–5 per lure

→

4

Extract RT

key_resp_9.rt or 3 + key_resp_8.rt

→

5

Compute Indices

REC = P(old|Target) − P(old|Foil)
LDI = P(sim|Lure) − P(sim|Foil)

Statistical Analysis & Multiple Comparison Control

Confirmatory + Exploratory framework with correction hierarchy

Tests Conducted

One-sample t-test vs 0: REC & LDI above chance per group

Independent t-tests: Between-group comparisons (3 pairs)

One-way ANOVA: Position effect (pre/mid/post) within groups

Paired t-tests: Pre vs post position per participant

Pearson r: REC–LDI correlation

Linear regression: LDI ~ lure bin (continuous)

Multiple Comparison Strategy

Bonferroni $\alpha = 0.0125$

Confirmatory H1–H4 (÷ 4 hypotheses)

Holm-Bonferroni **stepwise**

Post-hoc pairwise between groups

BH–FDR $q = 0.05$

Lure bin family (5 tests)

Uncorrected $\alpha = 0.05$

Exploratory H5–H7 (labelled)

03 Descriptive Statistics

Sample overview, response distributions, and preliminary findings

Sample Overview & Dataset Structure

160
Total Participants

24,000
Test Trials

9,600
Lure Trials

9,600
Target Trials

Group	N	Targets	Lures	Enc. Acc.
Item-Only	56	3,360	3,360	62.7%
Task-Only	53	3,180	3,180	62.4%
Both	51	3,060	3,060	61.0%

Response breakdown: old 37.9% · similar 34.5% · new 27.8%

Overall REC and LDI — Descriptive Statistics

All groups show robust memory well above chance

Group	REC M	REC SD	LDI M	LDI SD
Item-Only	0.634	0.153	0.319	0.169
Task-Only	0.596	0.174	**0.381**	0.188
Both	0.596	0.144	0.270	0.179

Task-Only shows significantly higher LDI than Both ($t=-3.08$, $p=.003$)

All groups: REC & LDI >> 0 (all $p < .001$, $d > 1.3$)

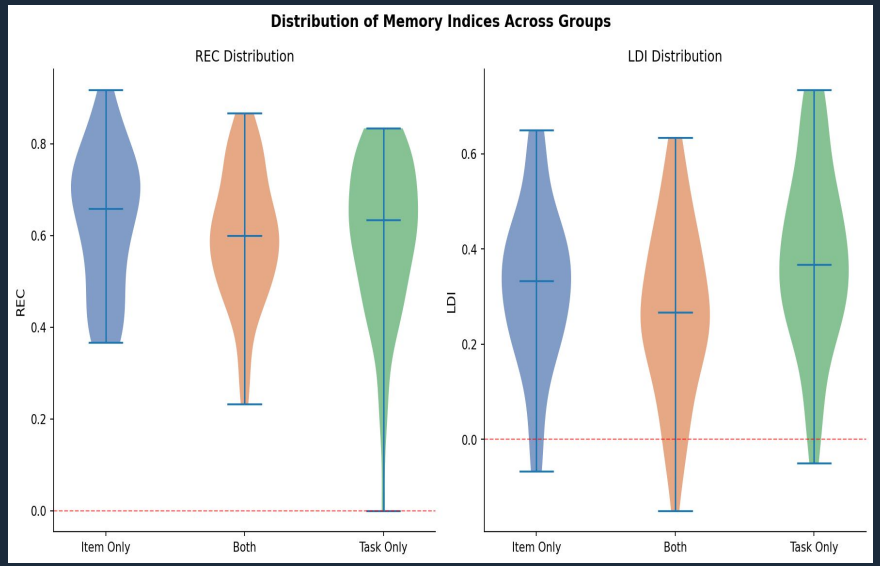


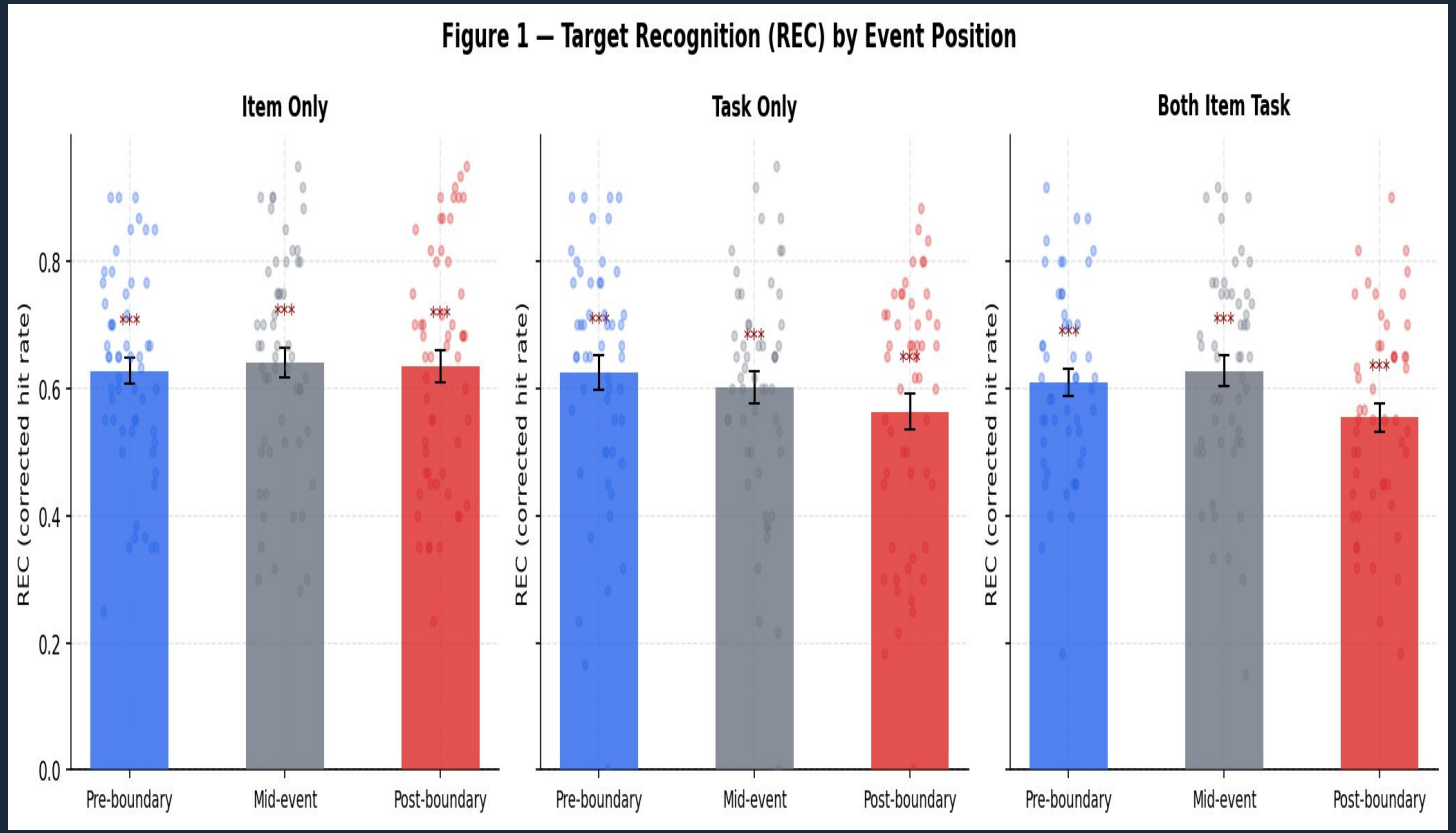
Fig. Distribution of REC and LDI by group

04 Results

Recognition memory · Lure discrimination · Position effects · Lure bins · RT

Figure 1 - Target Recognition (REC) by Event Position

All positions show strong recognition; post-boundary shows numerical decrease in Task-Only & Both



Key Stats

Item-Only:

No position effect

$F < 0.1$, $p = .93$

Task-Only:

Pre > Post trend

$t = 2.69$, $p = .010$

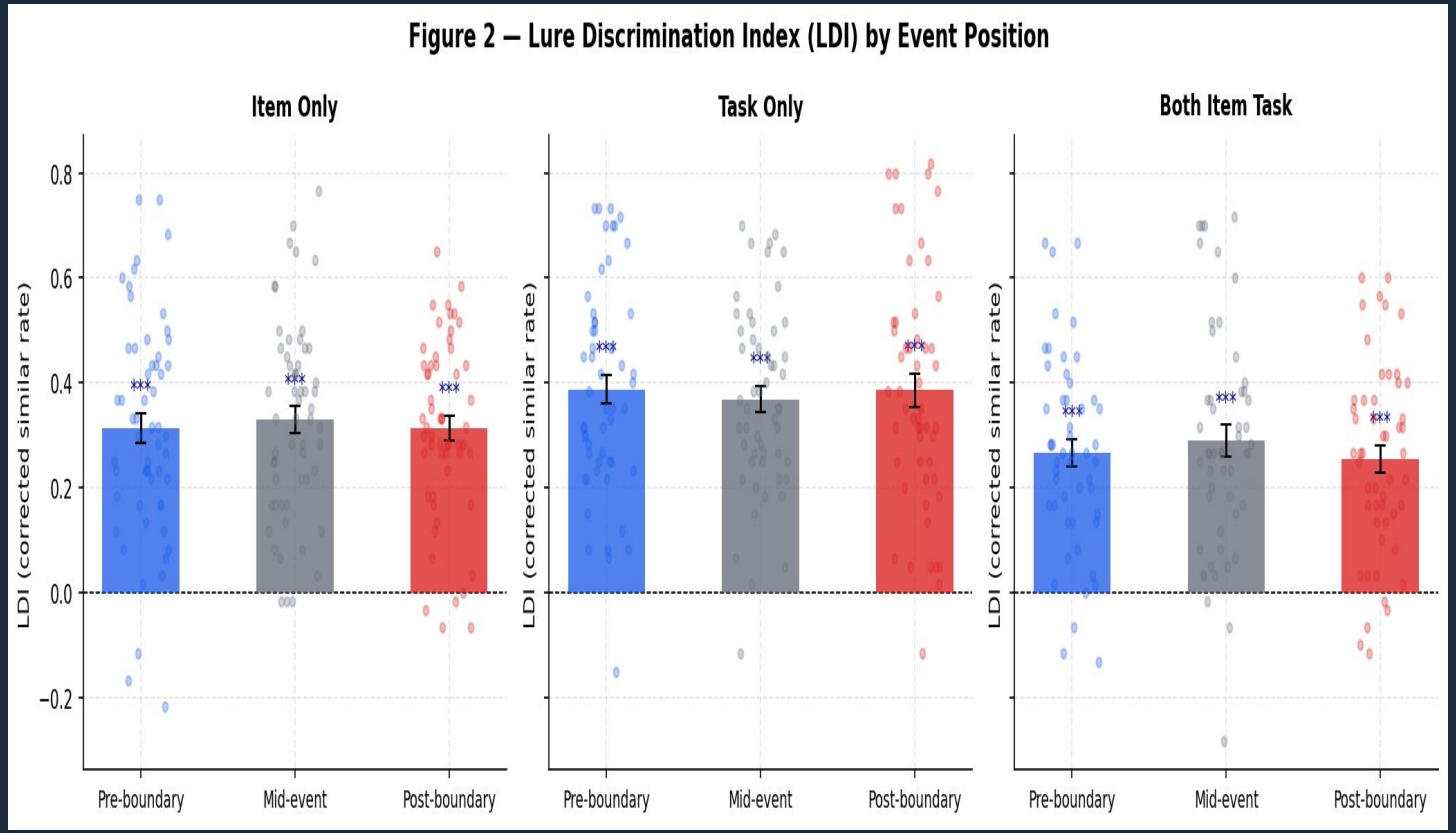
Both:

Mid > Post (sig)

$t = 3.48$, $p = .001$

Figure 2 - Lure Discrimination (LDI) by Event Position

Task-Only: balanced across positions; Both: slight post-boundary drop



Key Stats

LDI Pre-Post:

ANOVA across

all groups:

$F(2,157)=0.087$

$p = .917$ (ns)

All positions:

LDI >> 0

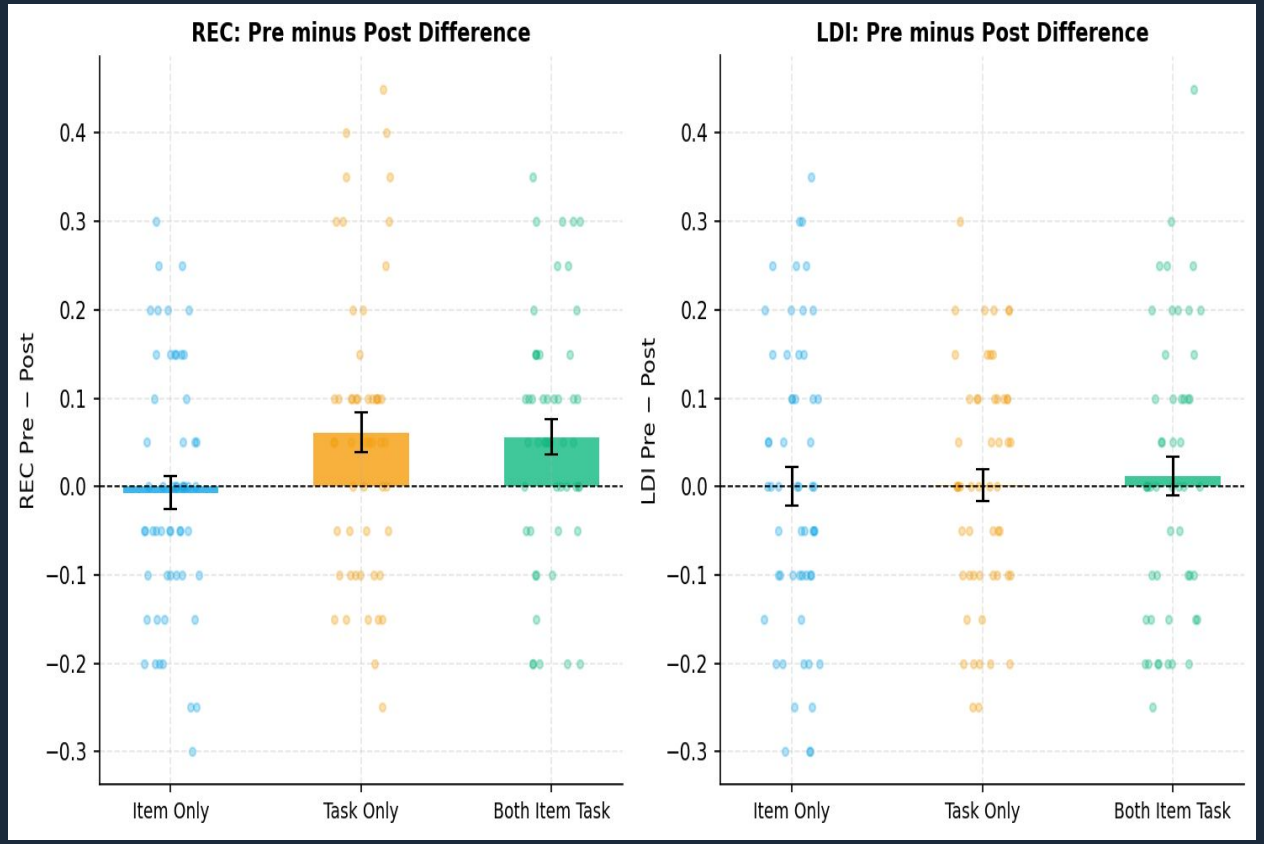
($d > 1.3$)

Task-Only LDI

highest overall

Pre - Post Boundary Effect: REC and LDI

Directional boundary effect on REC in Task-Only and Both; no effect on LDI



Interpretation

REC: Pre > Post

Task-Only: Δ = +0.061, t(52)=2.69, p=.010, d=0.37

Both: Δ = +0.056, t(50)=2.80, p=.007, d=0.39

Item-Only: Δ = −0.007 (ns)

ANOVA across groups:

F(2,157) = 3.52, p = .032

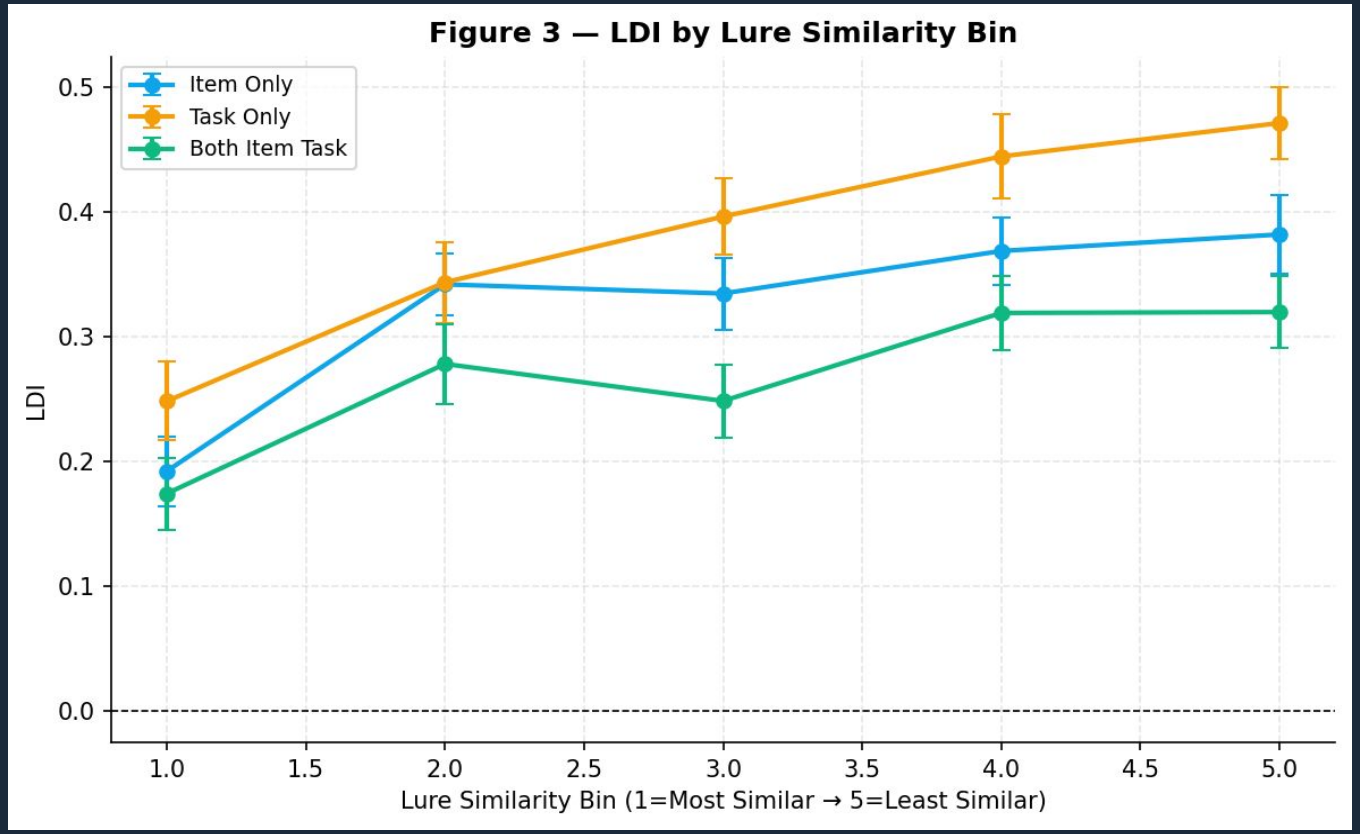
LDI: No position effect

F(2,157) = 0.087, p = .917

Boundary impairs rather than enhances recognition when encoding task is present

Figure 3 - LDI by Lure Similarity Bin

Monotonic increase confirms graded hippocampal pattern separation — H3



All bins sig. > 0 (BH-FDR corrected)

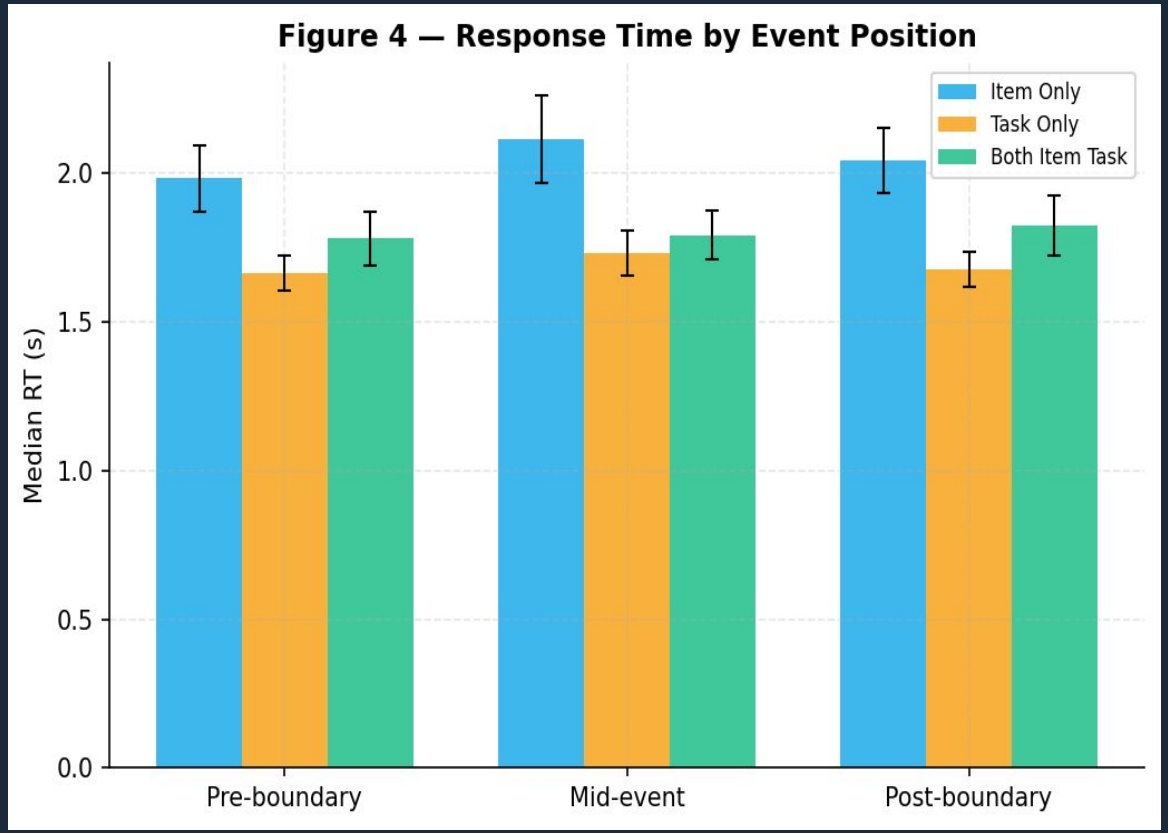
LDI Bin 1 -> Bin 5:

Item-Only:	0.19 -> 0.38
Task-Only:	0.25 -> 0.47
Both:	0.17 -> 0.32

Task-Only shows steepest gradient best discrimination

Figure 4 - Response Time by Event Position

Mid-event items elicit longest RTs — consistent with Event Segmentation Theory



RT (median, seconds)

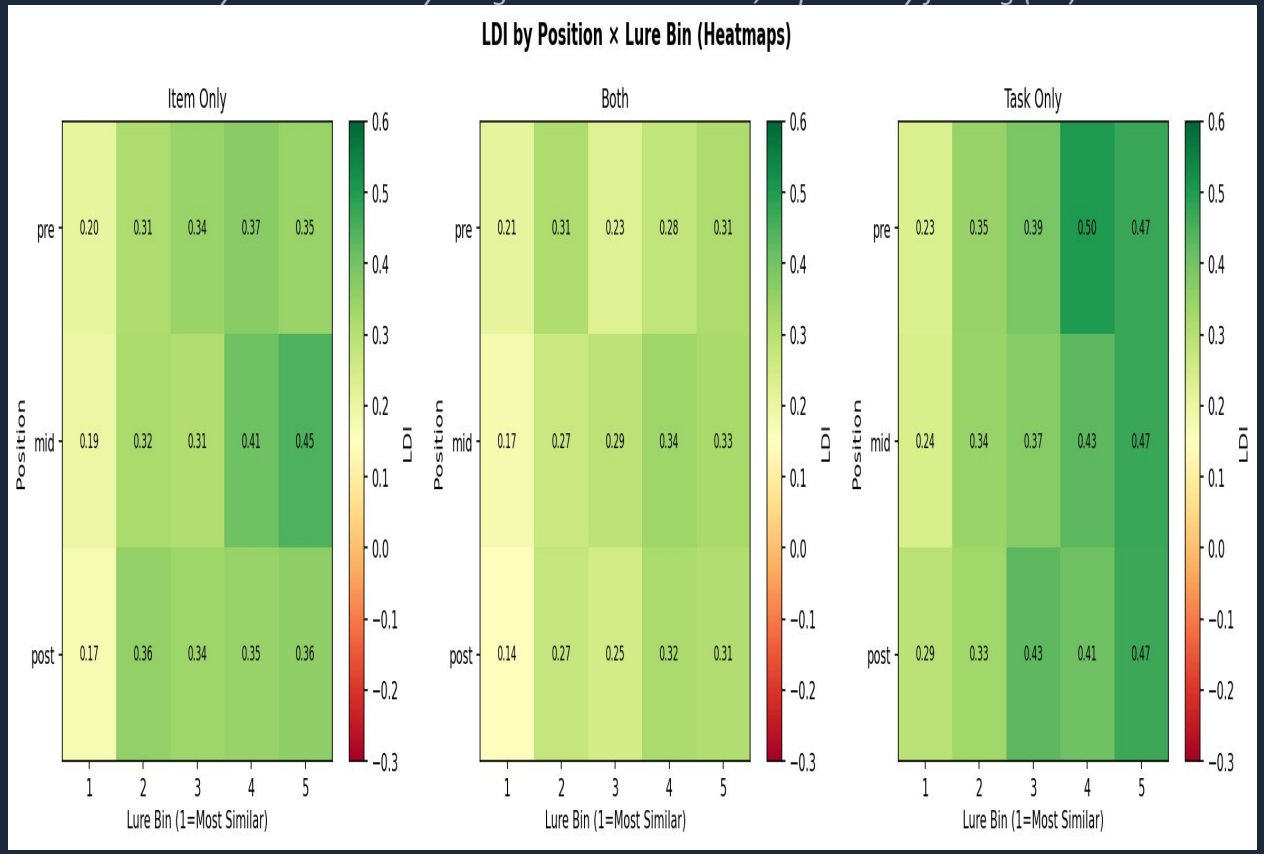
Group	Pre	Mid	Post
Item-Only	2.00	**2.13**	2.05
Task-Only	1.67	**1.74**	1.68
Both	1.81	**1.81**	1.84

EST prediction confirmed:
Mid-event > Pre consistently across all groups

Longer mid-event RT suggests greater cognitive engagement during stable event models

LDI: Position × Lure Bin Interaction

Post-boundary + low similarity = highest discrimination; exploratory finding (H7)



Reading the Heatmap

Green = high LDI
(better pattern separation)

Red = low LDI
(poor discrimination)

Rows = pre/mid/post
Cols = bins 1→5

Pattern: Bin 5 + post
consistently highest

Bin 1 (most similar)
challenges all positions

Between-Group Comparisons — All Statistical Tests

Key finding: Task-Only > Both for LDI (p = .003); no REC differences

Comparison	Metric	t-stat	p-value	Cohen's d	Result
Item-Only vs Both	REC	1.30	.195	0.25	ns
Item-Only vs Both	LDI	1.47	.146	0.28	ns
Item-Only vs Task-Only	REC	1.21	.229	0.23	ns
Item-Only vs Task-Only	LDI	-1.79	.076	0.34	ns (trend)
Both vs Task-Only	REC	0.04	.970	0.01	ns
Both vs Task-Only	LDI	-3.09	.003	0.61	** SIGNIFICANT

Bonferroni-corrected α = 0.0125 for confirmatory tests | LDI: Both vs Task-Only significant even after correction

05 Conclusions & Future Directions

Summary of findings and next steps

Conclusions

What we found and what it means

1 Memory above chance (H1)

REC ($M \approx 0.61$) and LDI ($M \approx 0.32$) significantly > 0 in all groups (all $p < .001$, $d > 1.3$). Shoebox encoding question produces robust recognition memory.

2 Boundary impairs REC in dual-task groups (H2 partially)

Pre > Post REC in Task-Only ($p = .010$) and Both ($p = .007$). Item-Only showed no boundary effect. Cognitive reset at boundaries may compete with encoding.

3 Lure similarity gradient confirmed (H3)

LDI monotonically increased from Bin 1 ⟶ Bin 5 in all groups (all BH-FDR corrected $p < .001$). Task-Only shows steepest gradient — best fine-grained discrimination.

4 Task-Only > Both for LDI (H4 partial)

Concurrent image viewing during encoding may interfere with item-specific encoding. Task-only semantic processing enhanced pattern separation.

Future Directions

Next steps to deepen understanding of boundary–memory interactions

Increase Statistical Power

Collect >20 trials per position/bin cell to detect small effects; use mixed-effects logistic regression treating items as random effects

Neuroimaging Integration

fMRI to measure hippocampal pattern separation signals at event boundaries; EEG for temporal dynamics

Delayed Testing

24-hour delay test to reveal whether boundaries differentially protect memories during overnight consolidation

Individual Differences

Trait anxiety, working memory capacity, and event segmentation ability as moderators of the boundary effect

Clinical Applications

Test MST boundary manipulation in MCI, depression, and stress cohorts as a sensitive hippocampal biomarker

Computational Modeling

Bayesian hierarchical models; signal detection theory; computational model of event-driven encoding

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Thank You

BRSM 2026 · MST & Event Segmentation Analysis

160	H1	H3	RT
Participants	Memory confirmed	Lure gradient	EST prediction